

Question:

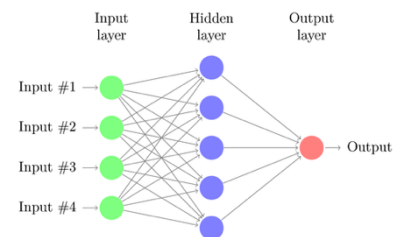
Describe components of a neural network and discuss the similarities of neural network with human brain.

What is a Neural Network?

A **neural network** is a model in Artificial Intelligence that is inspired by the structure and function of the **human brain**. It is used to recognize patterns, make decisions, and learn from data.

🧩 Components of a Neural Network:

Component	Description (Easy)
1. Neurons (Nodes)	Like brain cells. Each neuron receives input, processes it, and passes output.
2. Layers	Neurons are arranged in layers: ◆ Input Layer – takes input data ◆ Hidden Layers – process data using weights and activation functions ◆ Output Layer – gives final result
3. Weights	Numbers that decide how important each input is. They are adjusted during learning.
4. Bias	Extra value added to help the model make better decisions.
5. Activation Function	Decides whether a neuron should be activated or not (e.g., sigmoid, ReLU).



🧠 Similarities with the Human Brain:

Human Brain	Neural Network
Has neurons	Has artificial neurons (nodes)
Neurons are connected by synapses	Nodes are connected by weights
Learns from experience	Learns from data and training
Uses electrical signals	Uses numerical values (signals)
Brain adapts by strengthening connections	Neural networks adjust weights to learn

🎯 Example:

- Just like a child learns to recognize a cat by seeing many cats, a neural network learns to recognize cats by analyzing many cat images.

What does "learning" mean referring to neural nets?

◆ What is Learning in Neural Networks?

English:

In neural networks, **learning** means the process of improving the model's performance by **adjusting internal parameters** (called weights and biases) based on input data and expected output.

The goal is to **reduce the error** between the predicted output and the correct output.

বাংলায়:

Neural network-এ “learning” মানে হলো — **ওজন (weight) ও বায়াস (bias)** পরিবর্তন করে এমনভাবে শেখা, যাতে ইনপুট দিলে সঠিক আউটপুট পাওয়া যায়।
এই শেখার মাধ্যমে নেটওয়ার্ক **ধীরে ধীরে ভুল কমায়** এবং ভবিষ্যতে ভালো সিদ্ধান্ত নিতে শেখে।

◆ How does it learn?

1. **Input data** is given to the network
 2. It produces an **output (prediction)**
 3. The output is compared with the **correct answer**
 4. The **error** is calculated
 5. The network **adjusts weights** using algorithms like **backpropagation**
 6. This process repeats for many examples
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🎯 Example:

Suppose we want a neural network to recognize handwritten digits (0–9):

- We give it thousands of labeled images (e.g., image of “5” with label “5”)
 - At first, it guesses randomly
 - After each wrong guess, it **learns from the mistake** by adjusting weights
 - Over time, it becomes better at recognizing digits correctly
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Explain forward propagation and self-organization.

✓ 1 Forward Propagation (ফরওয়ার্ড প্রপাগেশন)

◆ Definition (English):

Forward propagation is the process in a neural network where **input data passes through the layers** (input → hidden → output) to produce a final output.

Each neuron:

- Receives input
- Multiplies it by **weights**
- Adds **bias**
- Applies an **activation function**
- Passes the result to the next layer
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$$Z = x_1 + w_1 + x_2 w_2 + \dots + b_a$$

◆ বাংলায়:

Forward propagation হলো এমন একটি ধাপ, যেখানে ইনপুট ডেটা ধাপে ধাপে **নেটওয়ার্কের প্রতিটি লেয়ার দিয়ে সামনে এগিয়ে যায়**, এবং শেষে একটি আউটপুট তৈরি হয়।

🎯 Example:

ধরুন, আপনি একটি ছবি দিলেন (ইনপুট)।

→ এটি প্রথমে ইনপুট লেয়ারে যাবে

→ তারপর হিডেন লেয়ারে প্রসেস হবে

→ শেষে আউটপুট লেয়ারে গিয়ে বলবে: “এটি একটি বিড়াল”।

✓ 2 Self-Organization (সেলফ-অর্গানাইজেশন)

◆ Definition (English):

Self-organization is a learning process where a neural network **automatically adjusts itself** without external supervision.

It finds **patterns or clusters** in data on its own.

This is used in **unsupervised learning**, like in **Self-Organizing Maps (SOM)**.

◆ বাংলায়:

Self-organization মানে হলো — নেটওয়ার্ক নিজে নিজেই শিখে নেয়, কোন ইনপুট ডেটা কীভাবে গ্রুপ করা যায় বা কোনটা কোনটার সাথে মিল আছে।

এখানে কোনো লেবেল বা সঠিক উত্তর দেওয়া হয় না।

🎯 Example:

একটি AI সিস্টেমকে অনেকগুলো ফলের ছবি দেওয়া হলো, কিন্তু বলা হয়নি কোনটা আপেল, কোনটা কলা।

→ AI নিজে নিজে ছবিগুলোর রঙ, আকার দেখে গ্রুপ করে নেয় — এটিই self-organization।

Describe the multilayer perceptron neural networks.

◆ What is a Multilayer Perceptron (MLP)?

English:

A **Multilayer Perceptron (MLP)** is a type of **artificial neural network** that consists of **multiple layers of neurons** — including an input layer, one or more hidden layers, and an output layer.

It is a **feedforward neural network**, meaning data flows in one direction:

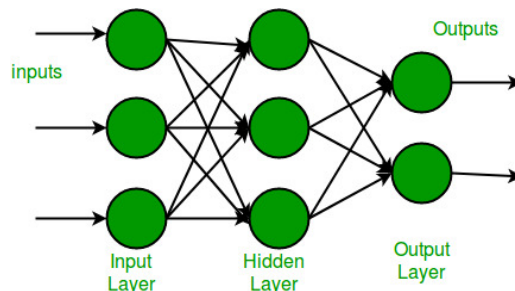
Input → Hidden Layers → Output

বাংলায়:

Multilayer Perceptron (MLP) হলো একটি কৃত্রিম নিউরাল নেটওয়ার্ক, যেখানে একাধিক লেয়ার থাকে —

ইনপুট লেয়ার, এক বা একাধিক **হিডেন লেয়ার**, এবং একটি **আউটপুট লেয়ার**।

এটি একটি **feedforward** নেটওয়ার্ক, অর্থাৎ ডেটা একদিকে প্রবাহিত হয় — পিছনে ফিরে যায় না।



◆ Structure of MLP:

1. **Input Layer:**
Takes raw data (e.g., image pixels, numbers)
 2. **Hidden Layers:**
One or more layers where neurons process data using **weights, biases, and activation functions** (like ReLU, sigmoid)
 3. **Output Layer:**
Produces the final result (e.g., class label, prediction)
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◆ Key Features:

- Uses **forward propagation** to compute output
 - Uses **backpropagation** to learn from errors and update weights
 - Can solve **non-linear** and **complex problems**
 - Works well for **classification, regression, pattern recognition**, etc.
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🎯 Example:

Suppose we want to recognize handwritten digits (0–9):

- **Input Layer:** takes pixel values of the image
 - **Hidden Layers:** extract features like curves, edges
 - **Output Layer:** gives the predicted digit (e.g., “5”)
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- Illustrate and describe the standard activation functions. **(2020)**
- How to adjust the weights of perceptron in neural network? **(2020)**

Khata